

Reptile Cardiology



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VET 437



UCDAVIS

VETERINARY MEDICINE

Optional (but recommended) reading

- Heart Disease in Reptiles: Diagnosis and Therapy by Lionel Schilling in Vet Clin Exot Anim 25 (2022) 383–407
- Mader's Reptile and Amphibian Medicine and Surgery chapter 68 on Cardiology

Learning Objectives

- Compare/contrast cardiac anatomy & blood flow of reptiles to mammals
- Identify unique features of cardiac-related diagnostics in reptiles
- Summarize clinical presentation of cardiac disease in reptiles
- Prioritize diagnostic tests based on clinical presentation in different reptile species
- Understand common etiologies of cardiac disease in reptiles

Lecture Outline

- Anatomy and physiology
- History and cardiovascular exam
- Diagnostics
- Etiologies of cardiovascular disease
- Treatment options

Reptile cardiac anatomy

- 3 (2) basic patterns

Squamates & Chelonians

Crocodilians

Varanids & Pythons

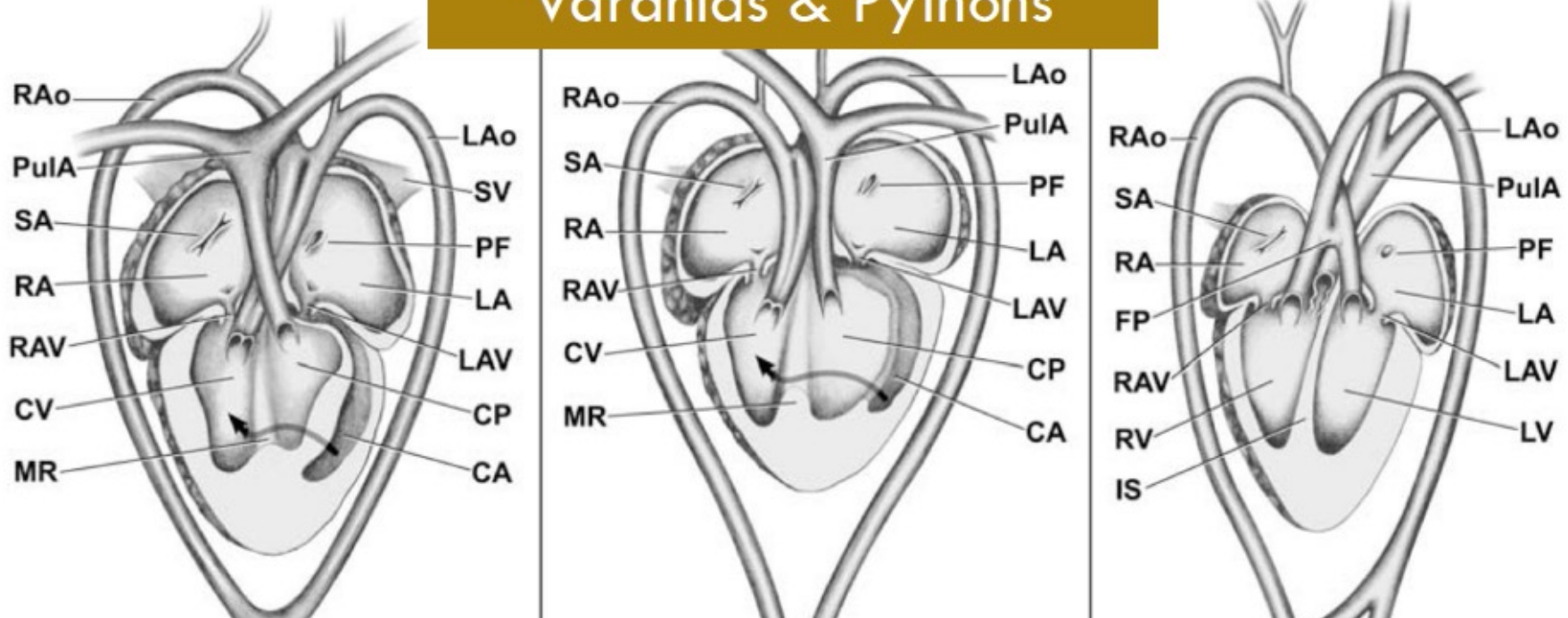
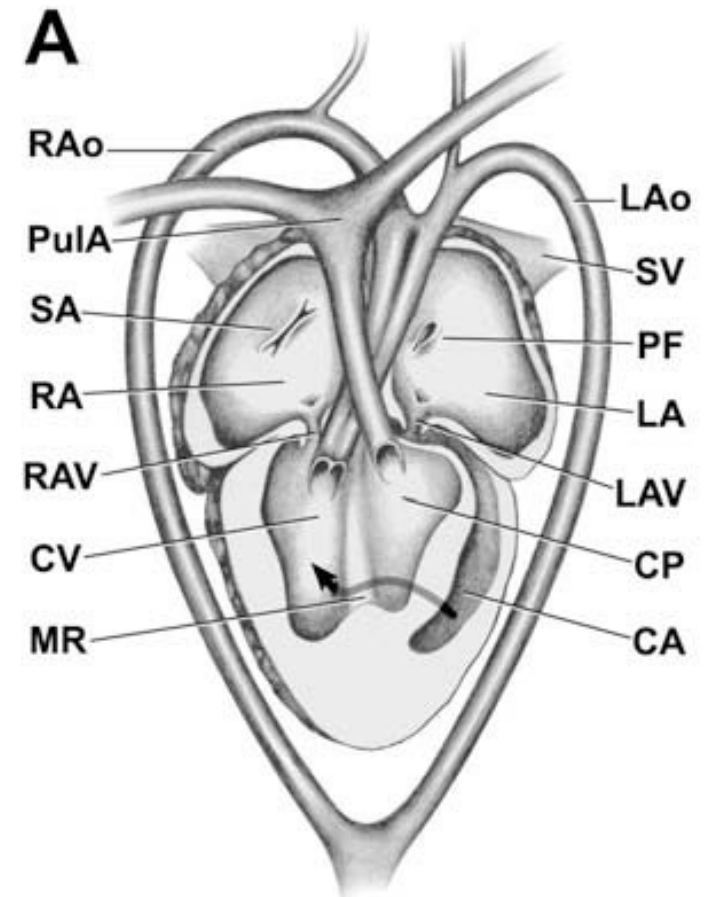


Fig. 1. Diagrams of reptilian heart structure, including chambers, compartments, and great vessels, comparing the turtles and general squamate pattern (A), varanid pattern (B), and that of crocodilians (C). The pulmonary artery (= pulmonary trunk) arises as a single structure then bifurcates. *Abbreviations:* CA, cavum arteriosum; CP, cavum pulmonale; CV, cavum venosum; FP, foramen of Panizza; IS, interventricular septum; LA, left atrium; LAo, left aorta; LAV, left atrioventricular valve; LV, left ventricle; MR, muscular ridge; PF, pulmonary vein foramen; PulA, pulmonary artery; RA, right atrium; RAo, right aorta; RAV, right atrioventricular valve; RV, right ventricle; SA, sinoatrial valve; SV, sinus venosus (Courtesy of Jeanette Wyneken, PhD, Boca Raton, FL).

Reptile cardiac anatomy

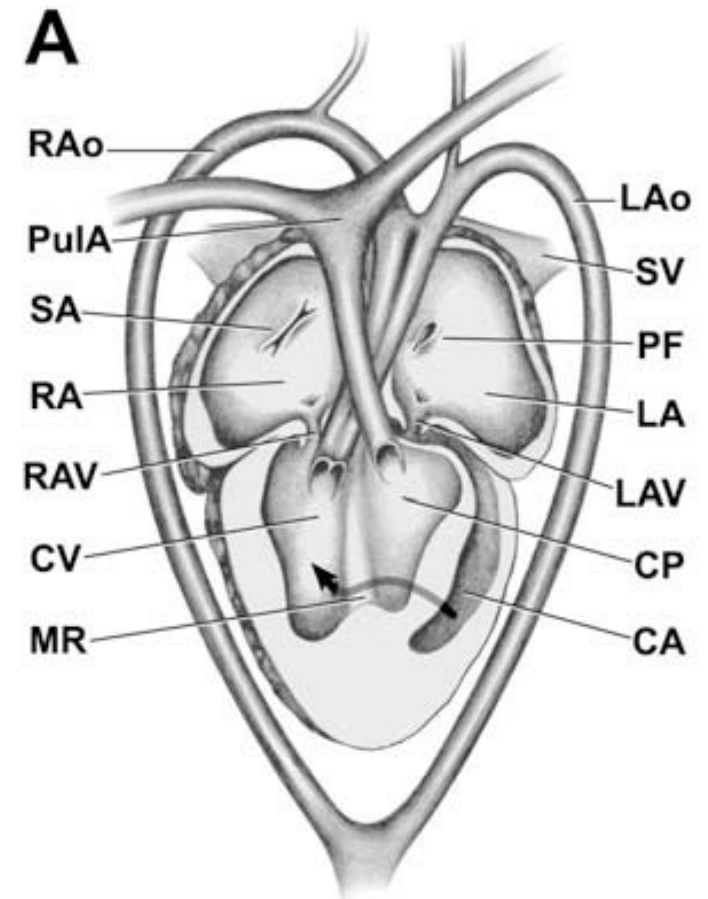
- General pattern
 - 3 “chambers” → 2 atria + 1 “complex ventricle”
- Anatomy 5-6 (sub)compartments
 - Sinus venosus (SV)
 - Right atrium (RA)
 - Left atrium (LA)
 - 2-3 ventricular chambers/compartments
 - Minimize mixing of deoxygenated/oxygenated blood
 - Muscular ridges, sequential contraction
 - AV valves
 - Systemic vs pulmonary arterial pressures
 - Varies by species/taxa & lifestyle demands

Squamates (lizards/snakes),
chelonians (turtles)



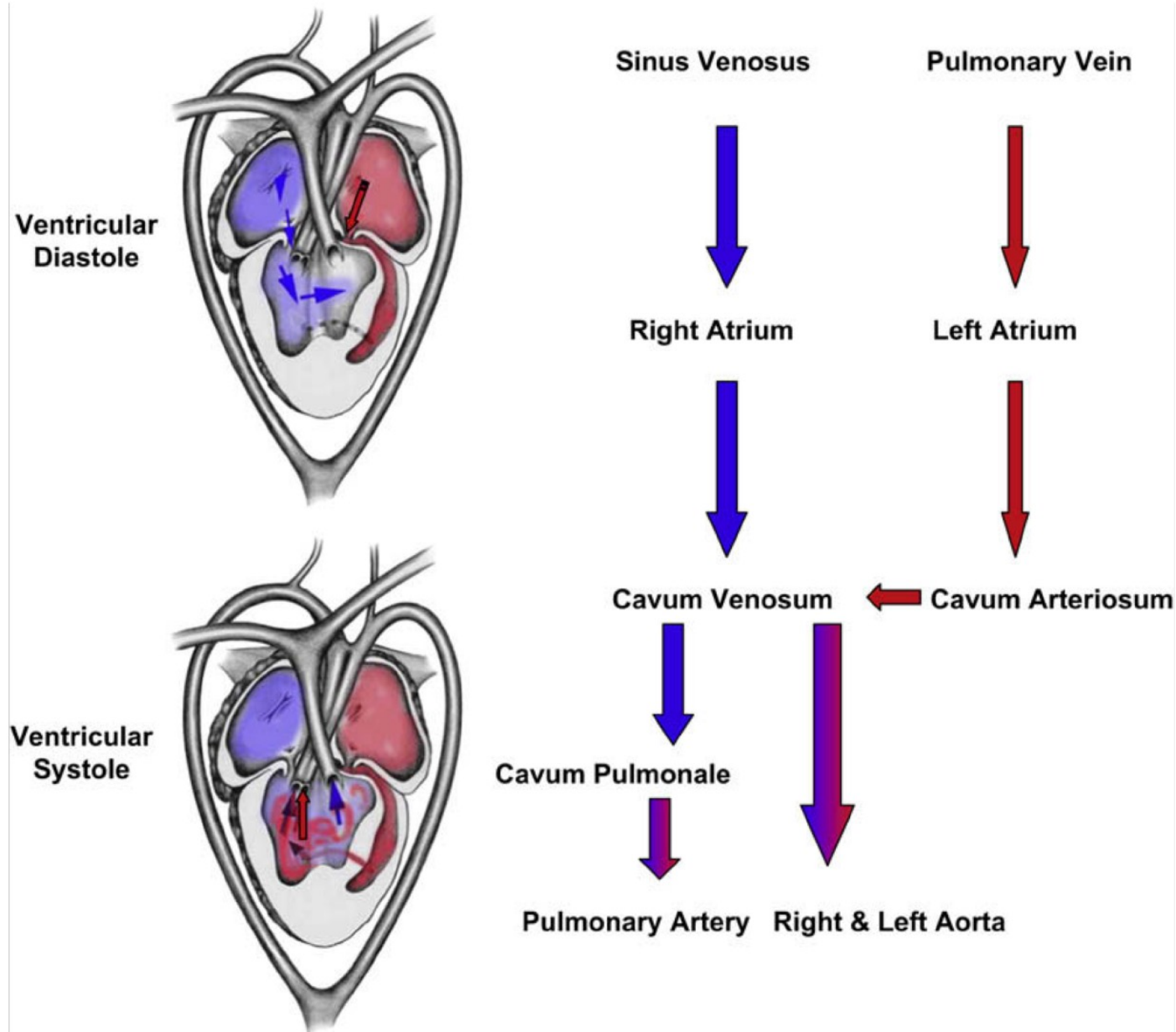
Reptile cardiac anatomy

- Ventricles (3 interconnected subchambers)
 - Cavum venosum (CV)
 - Cavum pulmonale (CP)
 - Cavum arteriosum (CA)
 - Muscular septum
 - Partially separates CV & CA
 - Incomplete
 - Functionally separates pulmonary & systemic blood flow
- Squamates (lizards/snakes),
chelonians (turtles)



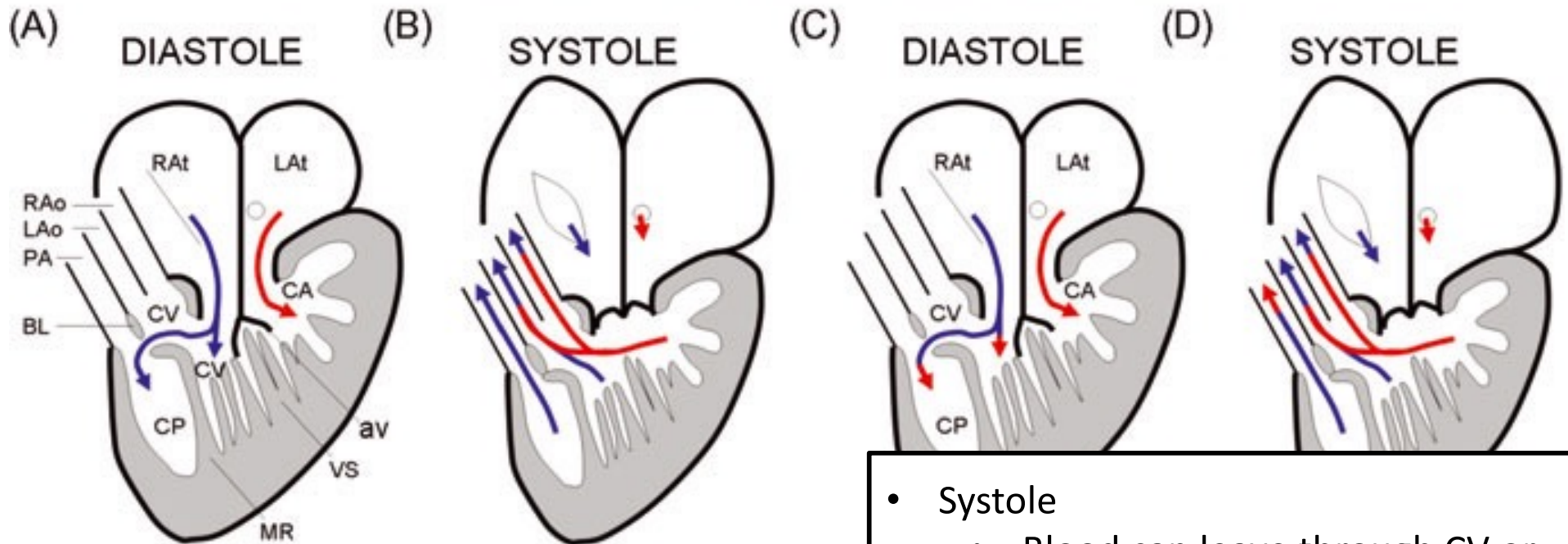
Reptile blood flow

Squamates (lizards/snakes), chelonians (turtles)



Squamates (lizards/snakes), chelonians (turtles)

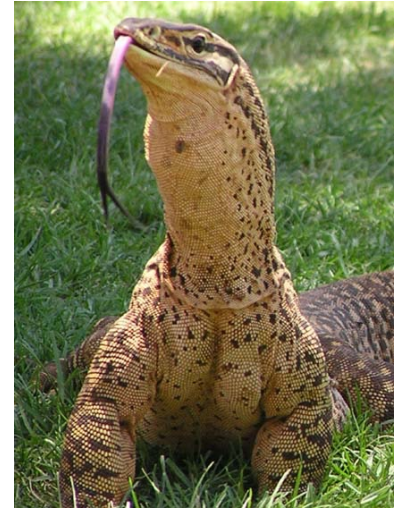
Squamates (lizards/snakes), chelonians (turtles)



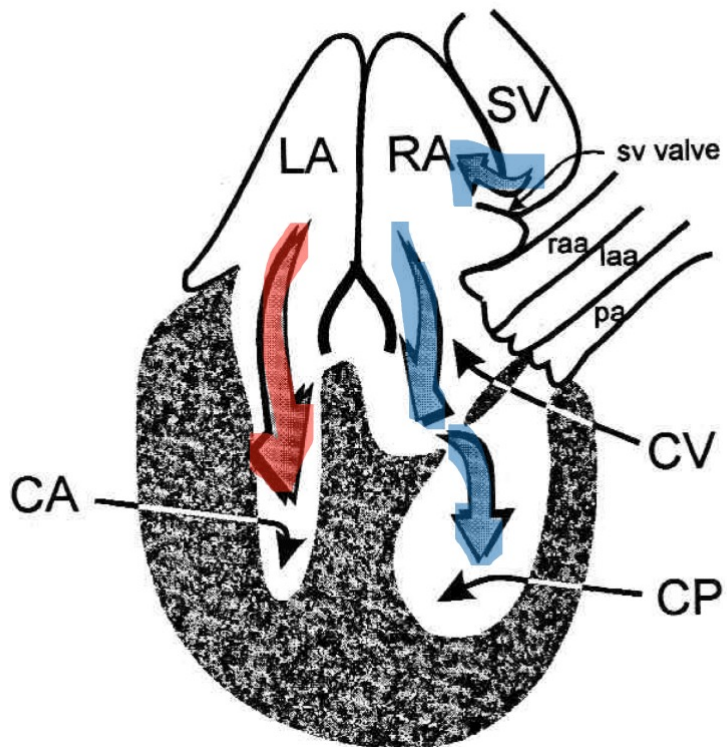
- Diastole:
 - Both atria supply ventricle with blood simultaneously
 - RA -> Cavum venosum (CV)
 - LA -> Cavum arteriosum (CA)
 - CV -> Cavum pulmonale (CP)
 - CV & CP continuous during diastole

- Systole
 - Blood can leave through CV or CP
 - Blood from CA -> interventricular canal to CV
 - CP -> pulmonary artery -> lungs
 - Can also enter LAo
 - CV -> RAo and LAo -> head/body
 - CV & CP separated by muscular ridge during systole

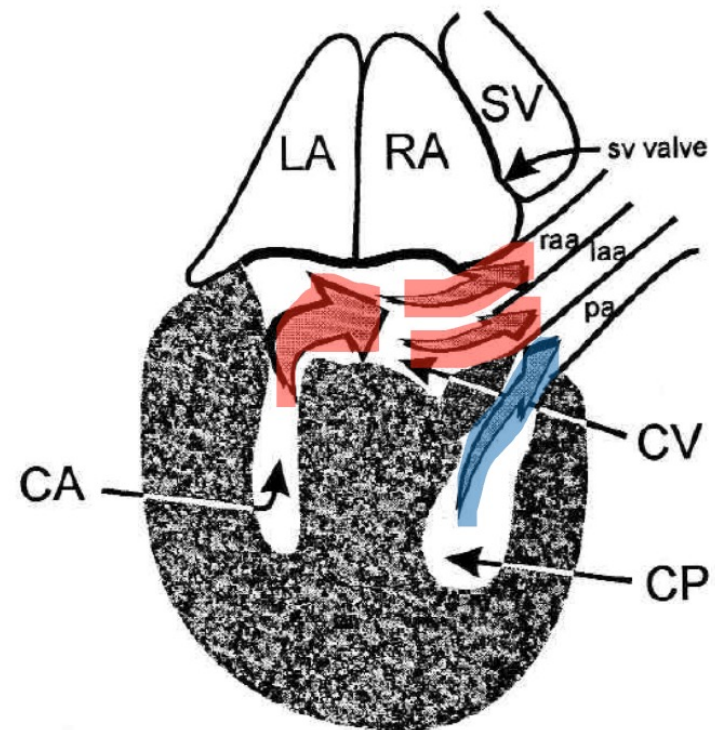
Varanids & pythons blood flow



Diastole



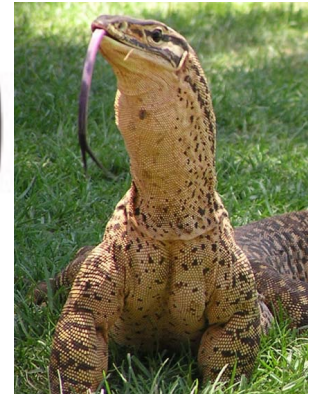
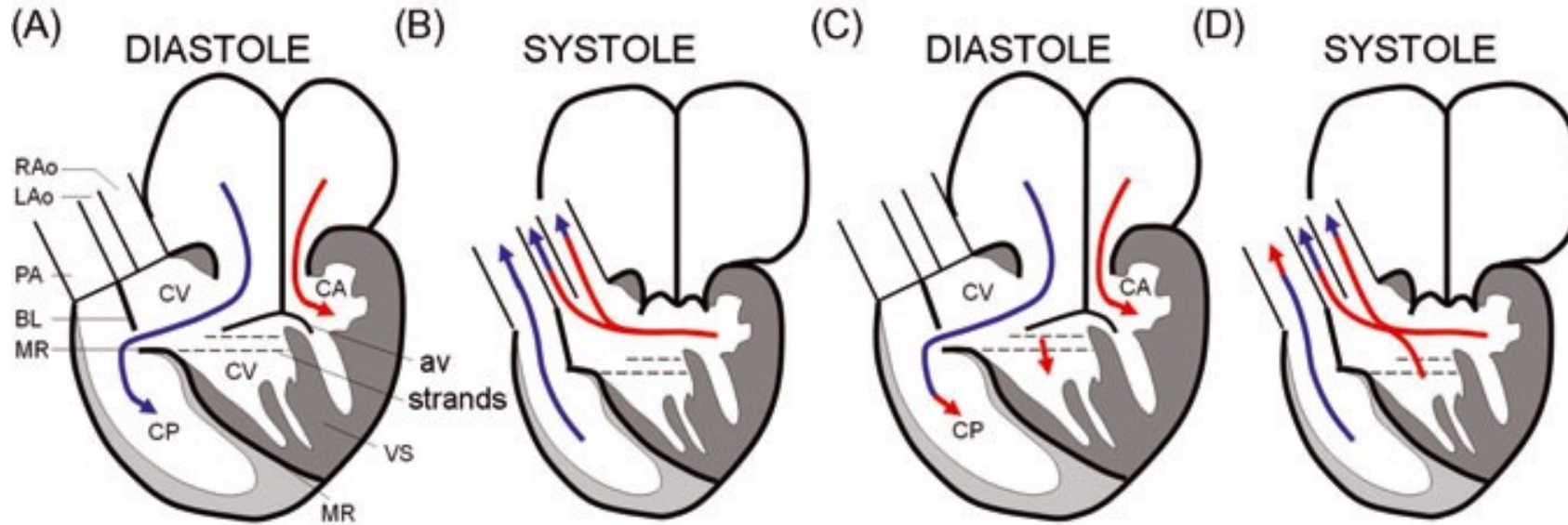
Systole



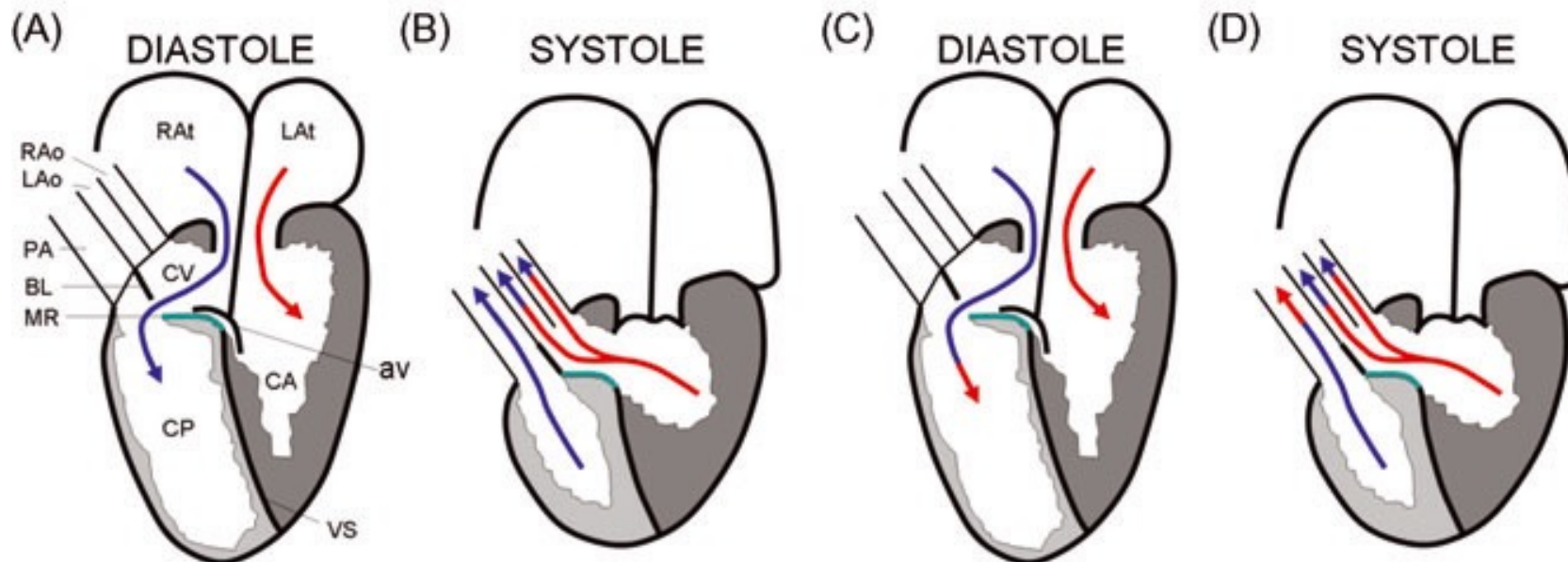
Structure and function of the hearts of lizards and snakes

Bjarke Jensen¹, Antoon F M Moorman, Tobias Wang

Varanid lizards



Pythons



BL = Bulbuslamelle
MR = muscular ridge
VS = vertical septum

Anaconda echo(!)

Review > [Biol Rev Camb Philos Soc](#). 2014 May;89(2):302-36. doi: 10.1111/brv.12056.

Epub 2013 Sep 2.

Structure and function of the hearts of lizards and snakes

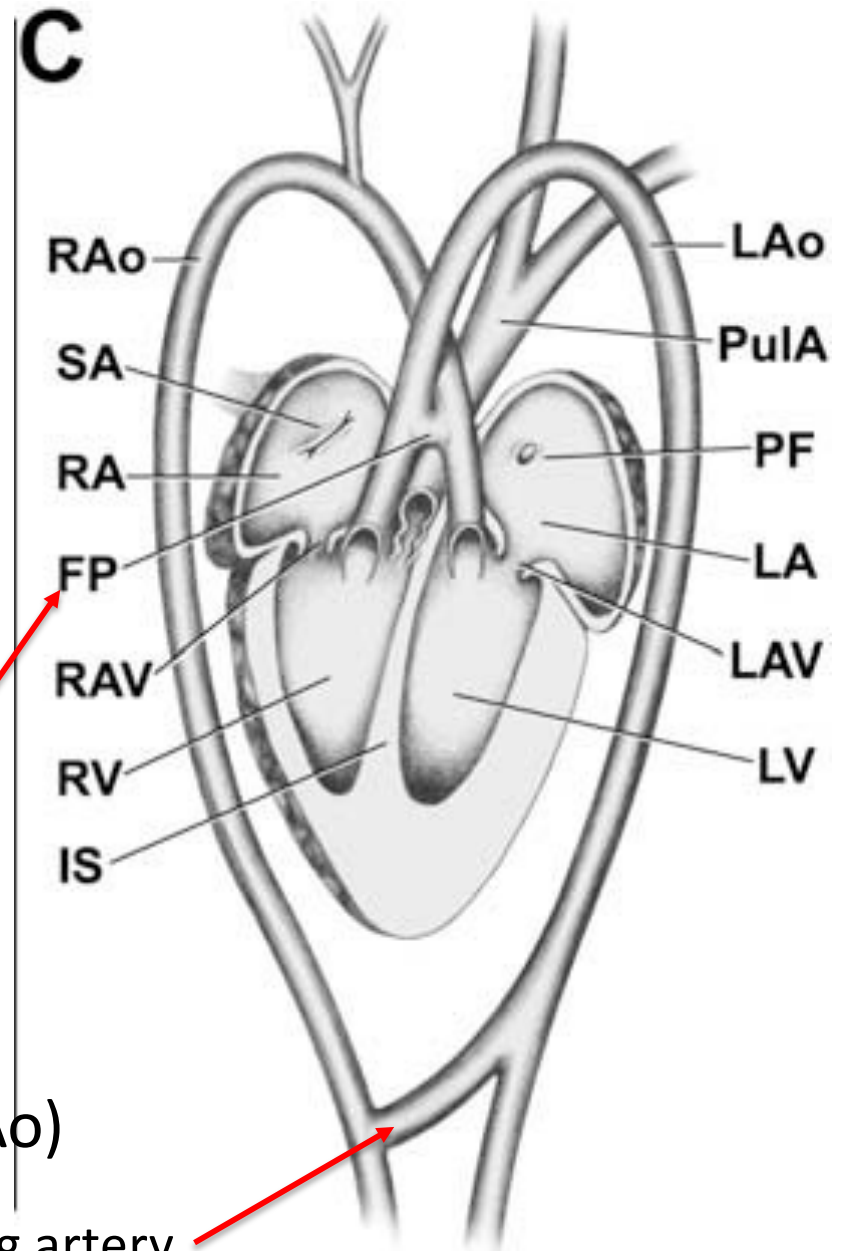
[Bjarke Jensen](#)¹, [Antoon F M Moorman](#), [Tobias Wang](#)

Agitated water (contrast) in
the systemic venous return
of a 6.2kg yellow anaconda

Crocodylians

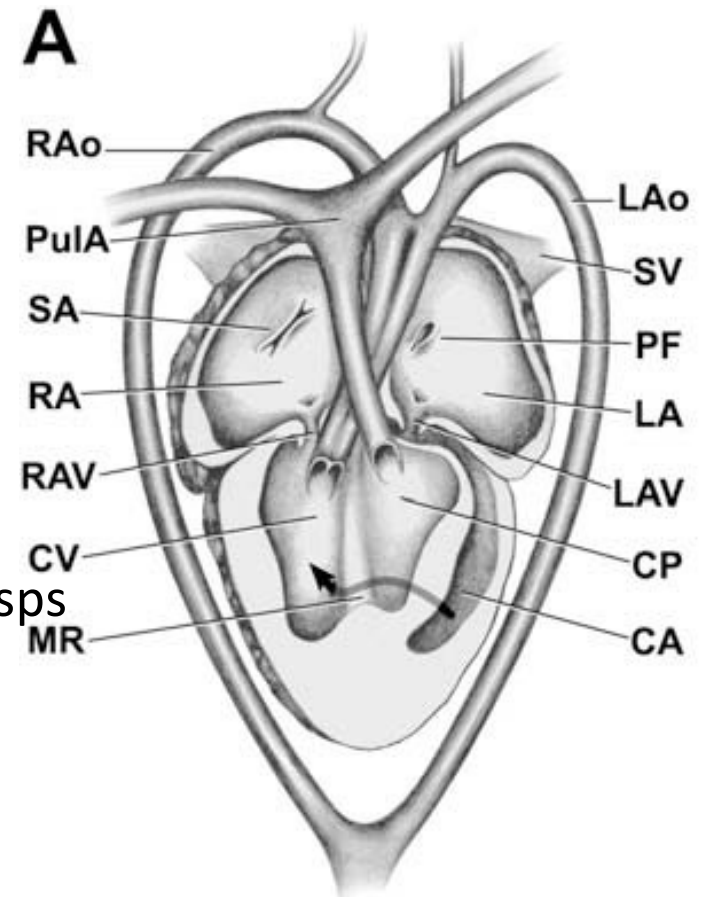


- Similar to mammals and birds
- Complete (fully developed) interventricular septum
 - RV (PA & LAo) & LV (RAo)
- 2 connected aortic arches (RAo, LAo)
 - Foramen of Panizza, dorsal connecting artery



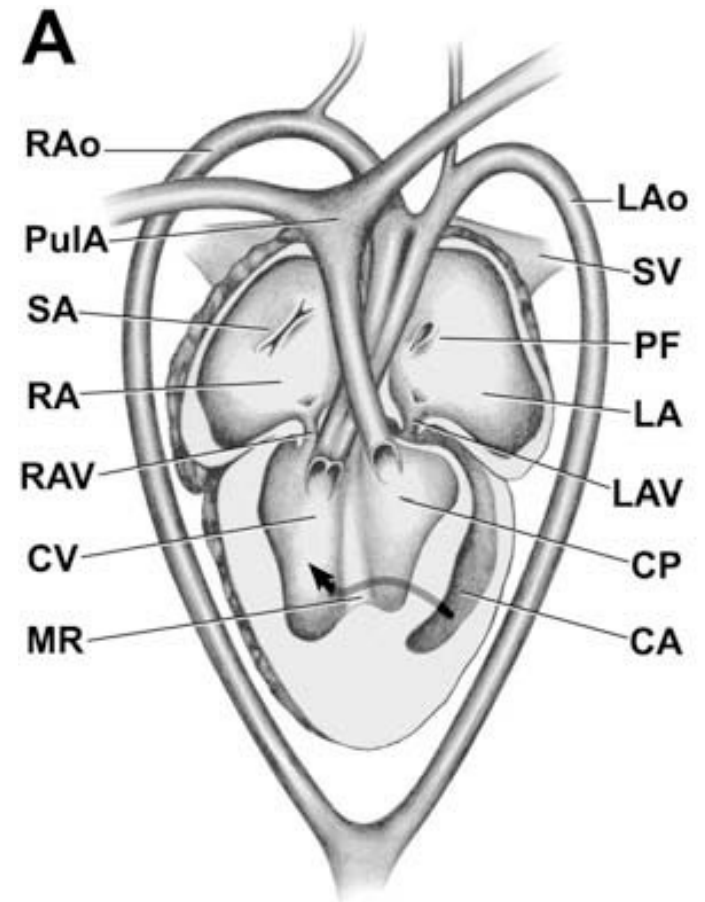
Cardiac valves

- Valves & ridges (“funnels”) separate sub-chambers and help direct blood flow
- Sinoatrial valve
 - Formed by flaps of endocardium
 - Separates SV from RA
- Atrioventricular valves
 - Variable: bell-shaped, single cusped, multiple cusps
 - Direct blood CV and CA during atrial systole
 - Prevent regurgitation during ventricular systole
- Aortae have bicuspid valves
- PA has semilunar valve



Great vessels

- 3 major (arterial) great vessels
 - Left aorta
 - Abdominal viscera then joins R-Ao
 - Right aorta
 - Head, gut, pancreas, spleen, duodenum then joins L-Ao
 - Pulmonary artery (PA)
 - RPA → right lung
 - LPA → left lung
 - Aortic base supported by cartilage, which can calcify with age



Reptilian electrophysiology

- Lack of specialized Purkinje fiber-based conduction system
- Contraction initiated in sinus venosus
- Sequential coordination of chamber contraction coordinated by myofiber arrangements within and around openings of chambers
- ***Systole (long), Diastole (short) -opposite of mammals***
- Heart rates are variable!
 - Lower than birds and mammals of similar size (↓metabolic rates)
 - Temperature (↑sun, ↓shade) → “heart rate hysteresis”
 - Activity (↑)
 - Respiration (↑ventilation, ↓apnea)
 - Digestion, pregnancy, handling stress

Lecture Outline

- Anatomy and physiology
- History and cardiovascular exam
- Diagnostics
- Etiologies of cardiovascular disease
- Treatment options

History & CV exam

- Signalment
 - Young → congenital heart disease
 - Young/middle age → cardiomyopathy, husbandry-related
 - Older → atherosclerosis (or mineralization of great vessels), degenerative valve disease, cardiomyopathy
- Activity / behavior
- Husbandry (environmental temperature, humidity, UV light, diet)
- Snake – swelling in cranial portion of body
- Often nonspecific findings
 - Decreased appetite, lethargy, open mouth breathing

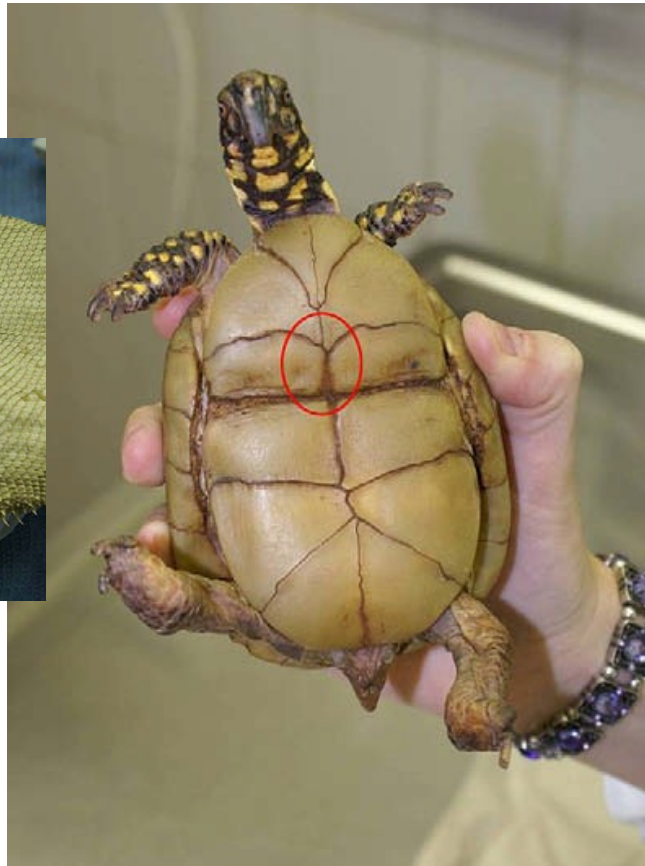
CV exam

- Heart location
- Variable → within coelomic cavity



Pectoral girdle

Possible to see heart
beating if abduct
forelimbs



Visualization impossible



Look for movement of
ventral scales

CV exam

- Mucous membrane color
 - Cloacal vent lining, oral
 - Cyanosis, ecchymosis may be seen
- Auscultation
 - Stethoscope (damp gauze between)
 - Doppler
 - Observation (snakes)
- HR varies with size, species, metabolic rate, season, time since last feed, temperature
 - Expected HR = $33.4 (Wt_{kg}^{-0.25})$ at optimal temp



CV exam

- Intraoral pulse can be felt in larger, sedated reptiles
- RR can vary with environmental changes and differences in activity
 - Obligate nasal breathers – open mouth breathing abnormal
- Swelling around level of heart
 - Snake heart can normally swell up to 40% in 48 hour after feeding
- Peripheral edema
- Coelomic effusion
- Organomegaly



Mader 2018

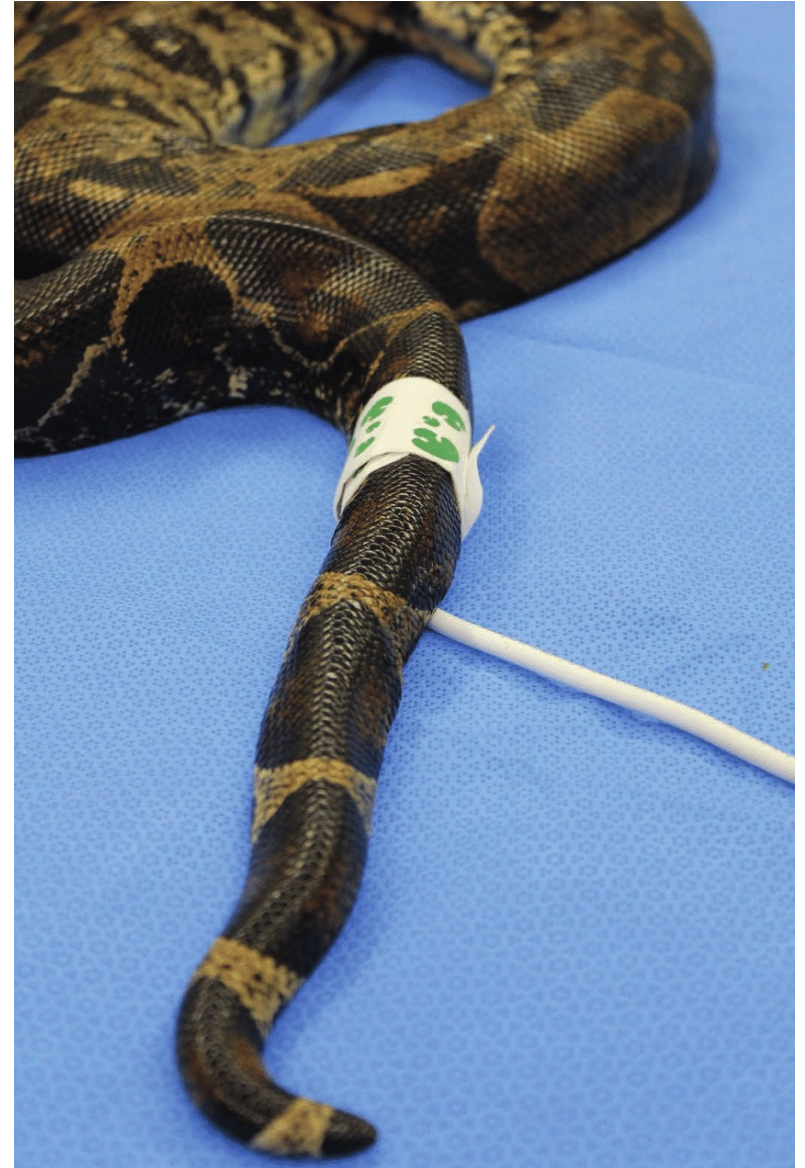
Lecture Outline

- Anatomy and physiology
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Blood pressure

Indirect blood pressure measurements

- Significant differences between invasive and noninvasive
- BP lower than comparably sized mammal or bird – varies by species
- Reference ranges available for multiple species



Blood work (biochemistry panel)

- Hydration → albumin concentration
- Creatinine kinase (CK)
 - Release from damaged myocytes, including cardiomyocytes
 - Green iguanas: highly correlated with cardiac muscle
 - LDH also released from damaged cardiomyocytes
- Cardiac troponin-I – not evaluated in reptiles
- Minerals/Electrolytes (Ca, Phos, Na, K)
 - Might affect myocardial function and cardiac conduction/rhythm
- Hyperuricemia → myocardial gout
- Cholesterol levels – hypercholesterolemia associated with atherosclerosis in a bearded dragon

ECG

- Legged reptiles similar to mammals, neck if poor forelimb tracings
- Snakes
 - Electrodes two heart lengths cranial and caudal to heart
 - Several placement strategies described
- AliveCor is an alternative

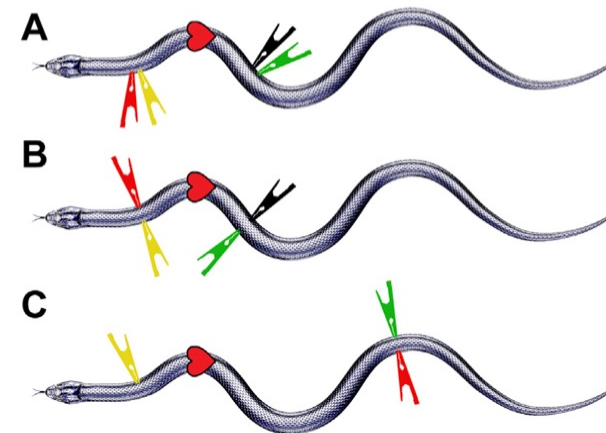
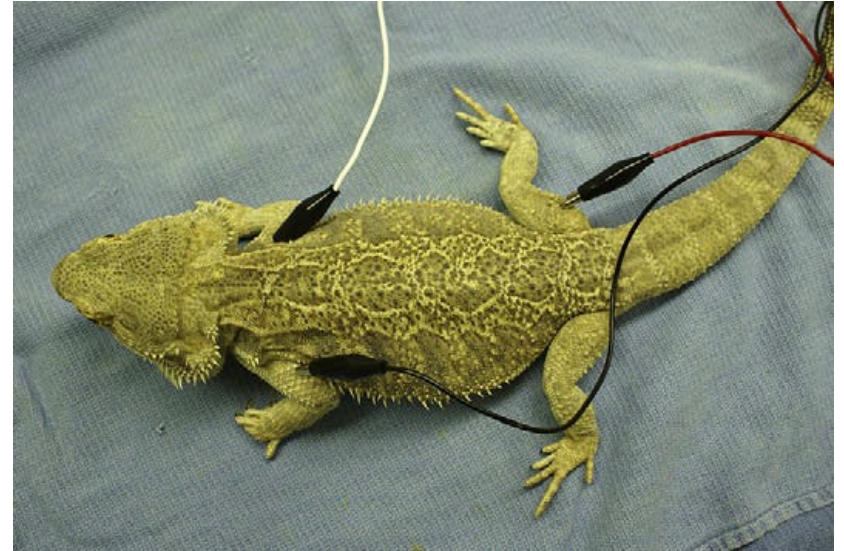


Fig. 8. ECG electrode placement in snakes. (A) According to Karthik M, Rajkumar K, Krishna MK, et al, 2013. (B) According to Kharin S and Shmakov D, 2009. (C) According to Mitchell MA, 2009 and Bogan JE, 2017. Figure illustrated by Pierre Morice.

ECG

- Waves: (SV) P, QRS, T
- Long systole (QT long) , short diastole (TP short)
- Reference ranges exist for some species, can establish own “normals”
- Low voltages are typical

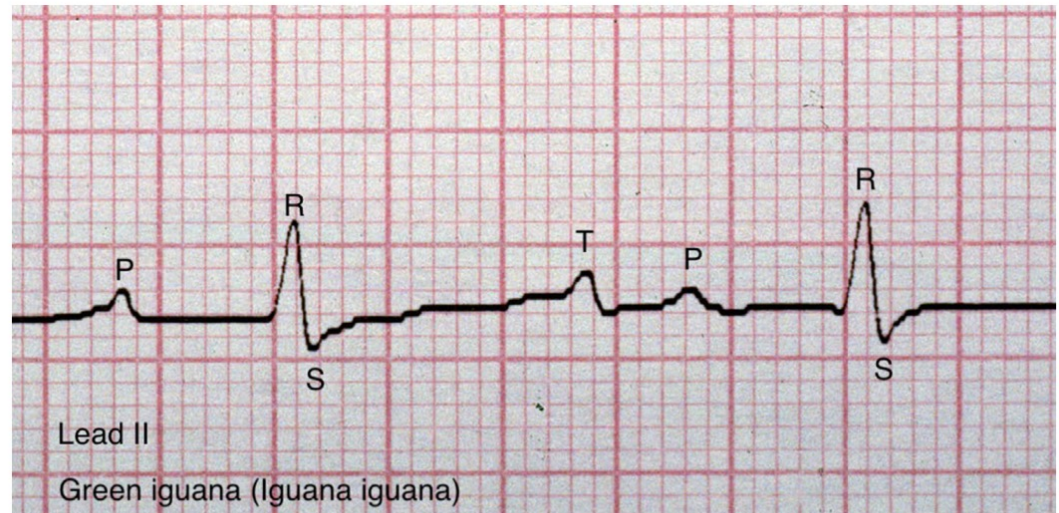


Fig. 13. Typical ECG complex in reptiles. SV wave is absent. P, P wave; R, R wave; T, T wave.

Table 14-2 Evaluation of Reptile Electrocardiographic Results

Heart rate	Use standard techniques (at 50 mm / s, 1 mm = 0.02 s)
Identify components	P-wave, QRS-complex, T-wave, SV-wave (if present)
Rhythm	P-wave for each QRS? / QRS for each P-wave?
Evaluate components Normal for species?	Intervals uniform in length? Consider temperature and heart rate differences
Integrate into clinical setting	Compare with “normal” control patient Is it consistent with patient findings and ancillary examinations?

ECG

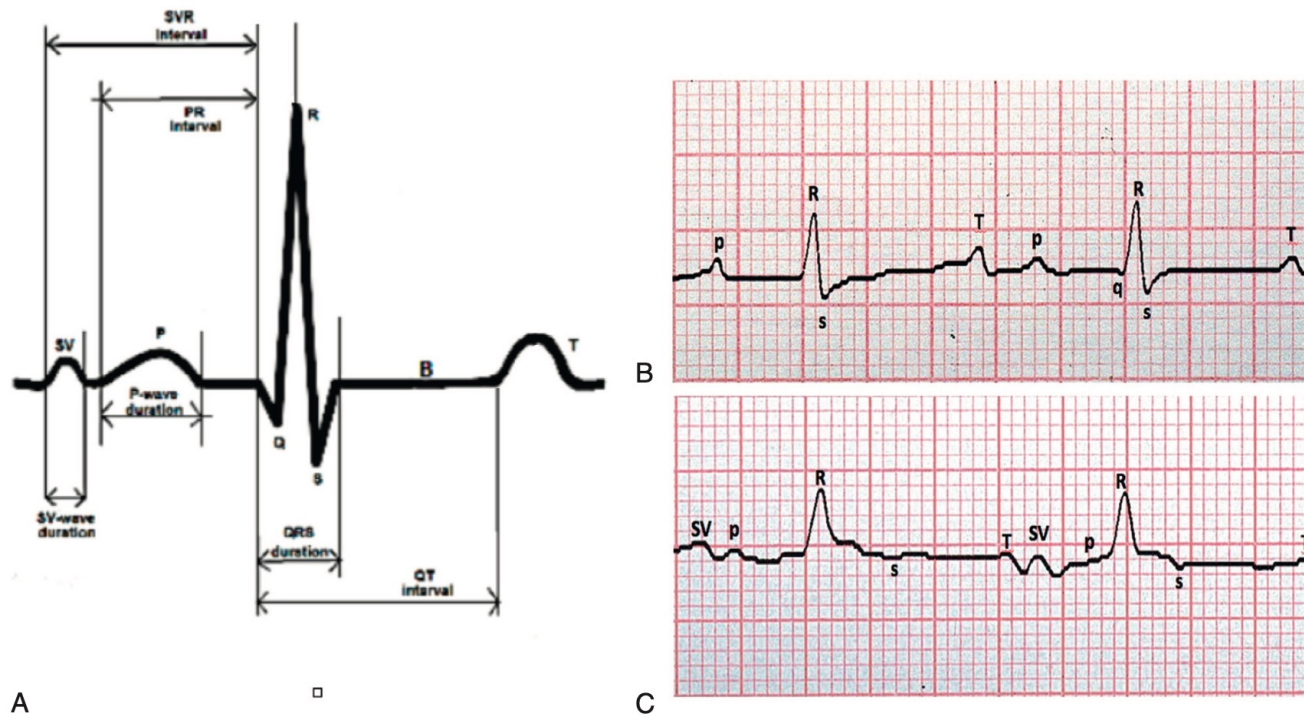


FIG 68.16 (A) Typical ECG complex in reptiles. B, Baseline. P, P wave; QRS, QRS wave; SV, SV wave; T, T wave. (B) ECG tracings in a green iguana (*Iguana iguana*). (C) ECG tracings in a Dumeril's Madagascar boa (*Acrotaphus dumerilii*). (A, Courtesy of Craig Hunt; B and C, courtesy of Lionel Schilliger.)

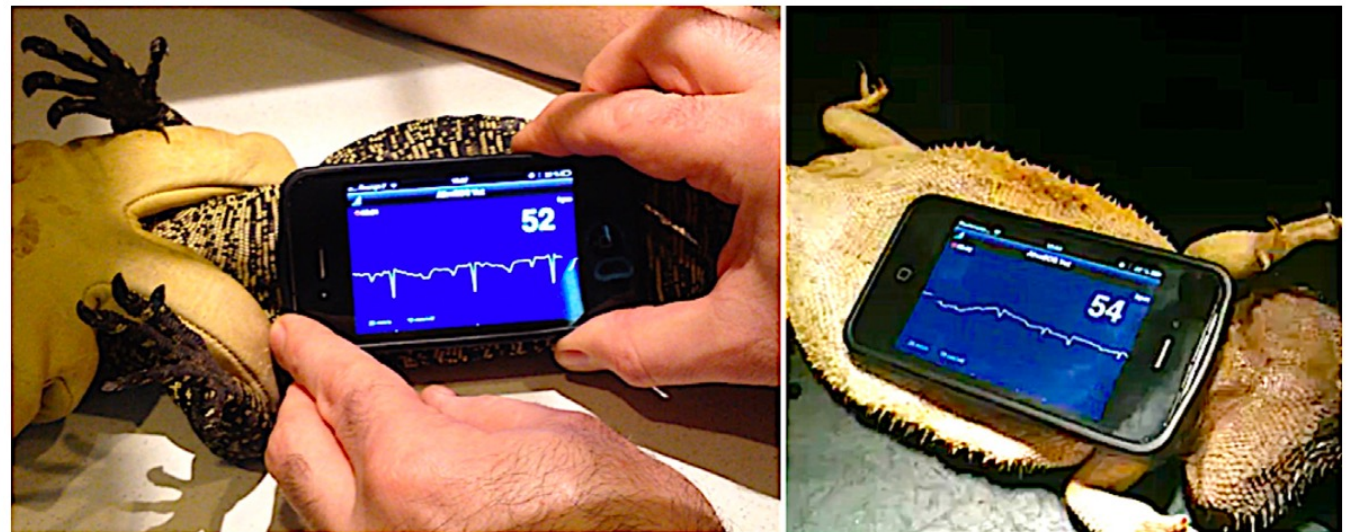


Fig. 12. Use of a single-lead ECG recorder (AliveCor Veterinary Heart Monitor®) connected to a smartphone case in a varanid (*Varanus indicus*) (left) and in a bearded dragon (*Pogona vitticeps*) (right).

Radiography

- Compare to known normal (clinical experience)
- Some lizards difficult to visualize cardiac silhouette because of bony pectoral girdle. Chelonians and crocodilians very challenging
 - Varanids exception due to more caudal location of heart
- Pulmonary edema, hepatomegaly, vessel changes



Radiography

Varanid

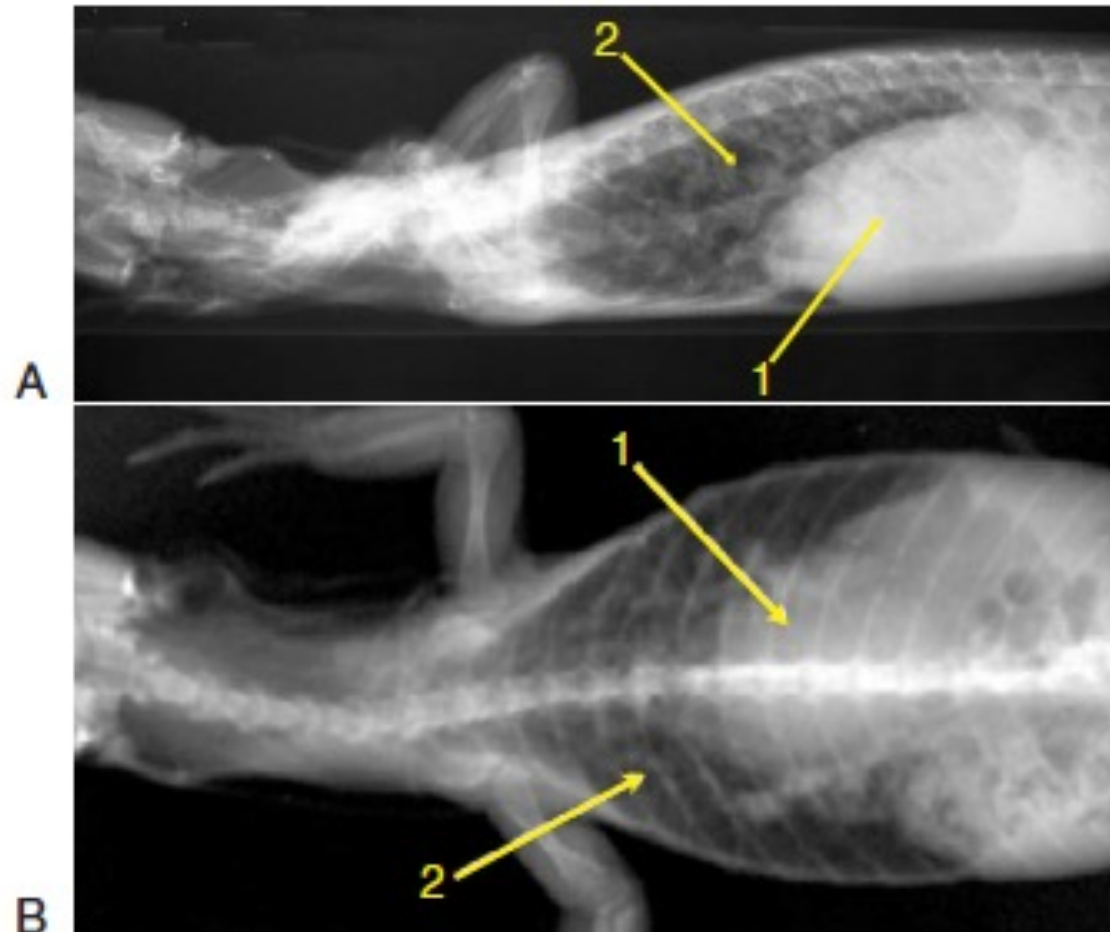
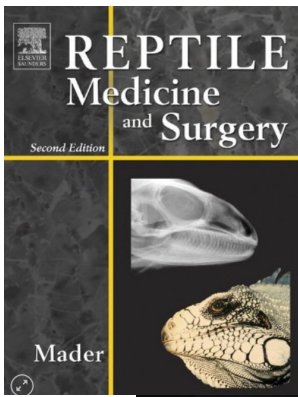
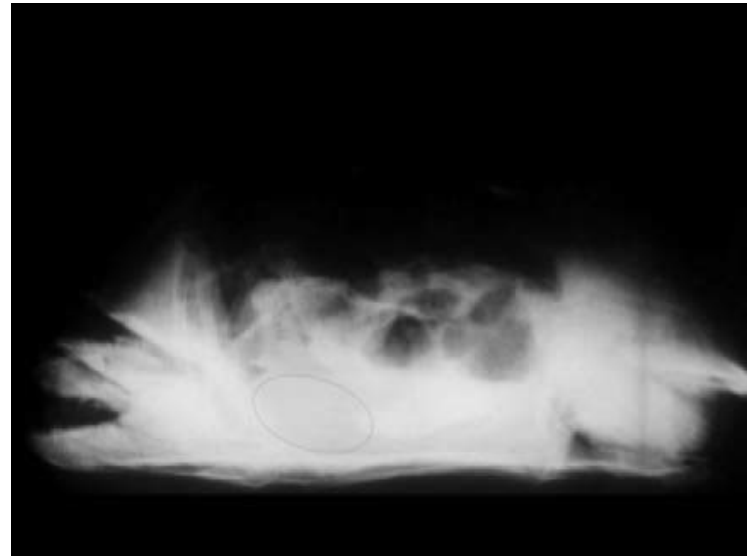
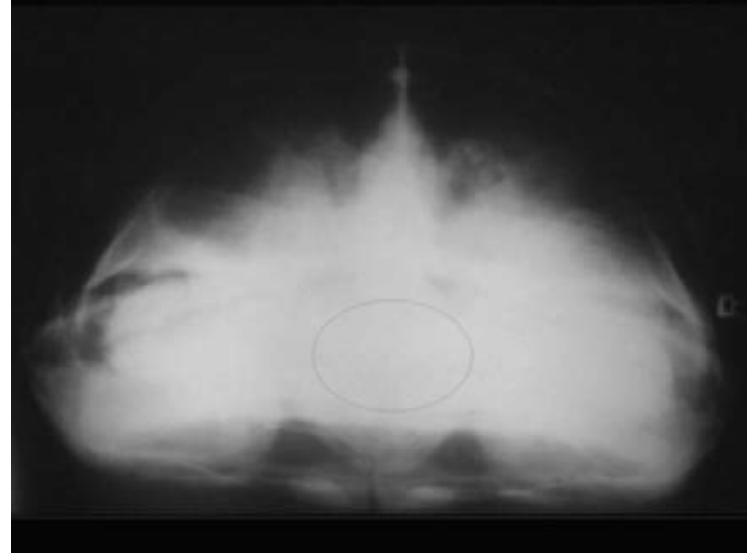


FIGURE 14-1 Swelling in the area of the heart, pleural and peripheral edema, ascites, cyanosis, and ecchymoses are all clinical signs that suggest cardiovascular disease. This monitor presented with weakness, obvious cardiomegaly, and ascites. Marked cardiomegaly was noted radiographically (1), as well as pulmonary edema (2). (Photographs courtesy B. Okimoto.)



Radiography



Radiography

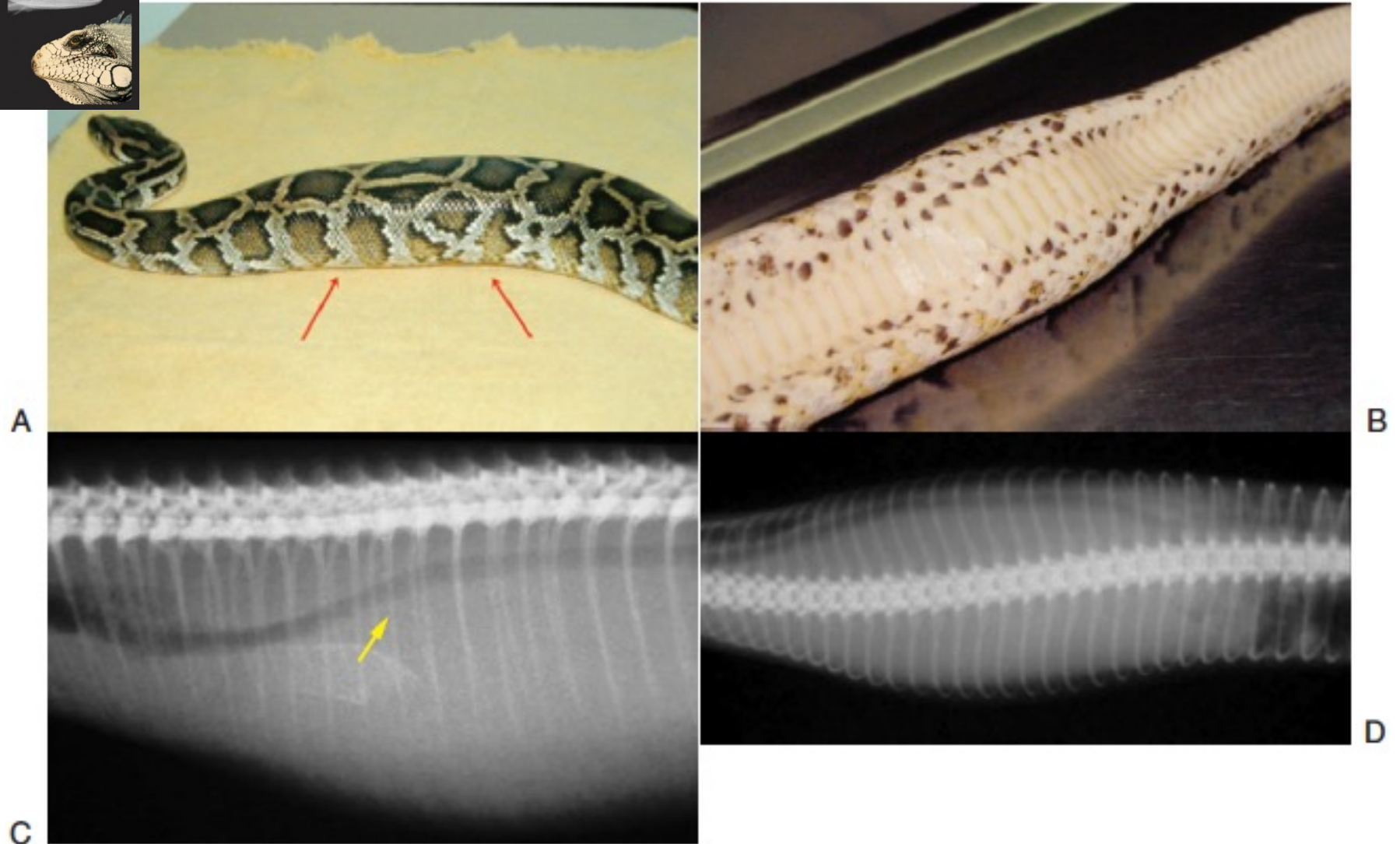
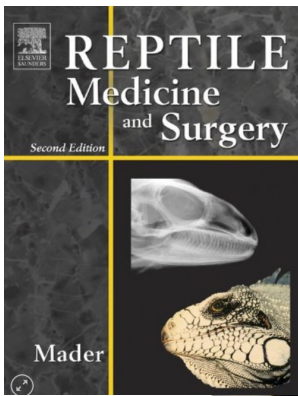
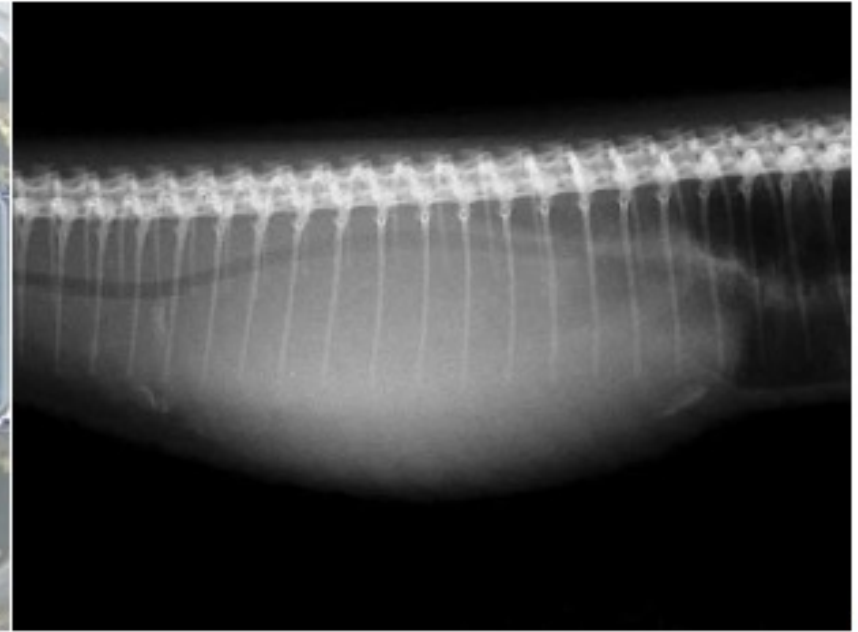


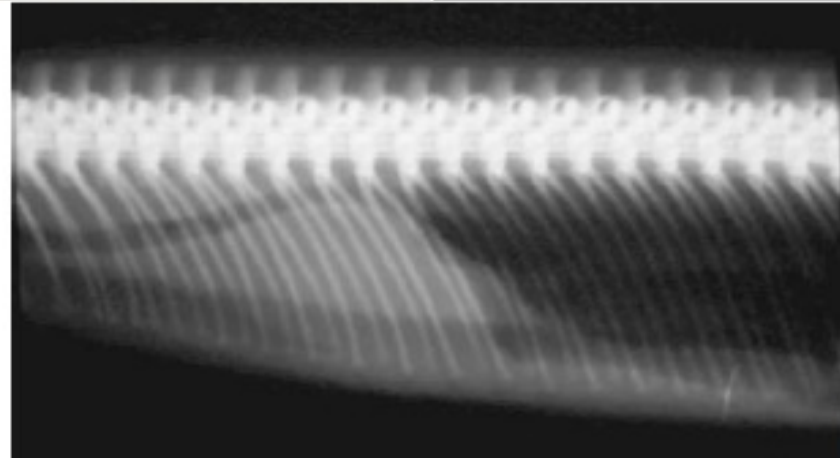
FIGURE 14-9 A and B, Burmese Python (*Python molurus bivittatus*) with cardiomegaly (between arrows). C and D, Radiographs of the patient show calcification at the base of the aorta (arrow). Aortic stenosis resulted in ventricular hypertrophy. (Photographs courtesy D. Mader.)



Radiography



B



C

FIGURE 14-8 A, This Carpet Python (*Morelia spilotes variegata*) was seen for swelling in the region of the heart. B, Radiograph of the patient shows an enlarged heart. C, Radiograph from a normal python for comparison. As diagnostics and therapeutics become more sophisticated, these cases will no longer carry a grave prognosis. (A and B, Photographs courtesy S. Barten; C, Photograph courtesy D. Mader.)

Echocardiography

- Challenging!
- Axial for lizards
- Ventral/left lateral for snakes
- Cervicobrachial for chelonians
- Air trapping b/w scales
 - Gel
 - Water bath



Fig. 22. Placement of an ultrasound probe (lateral approach, left axillary ultrasound window) in a green iguana (*Iguana iguana*) in sternal recumbency.

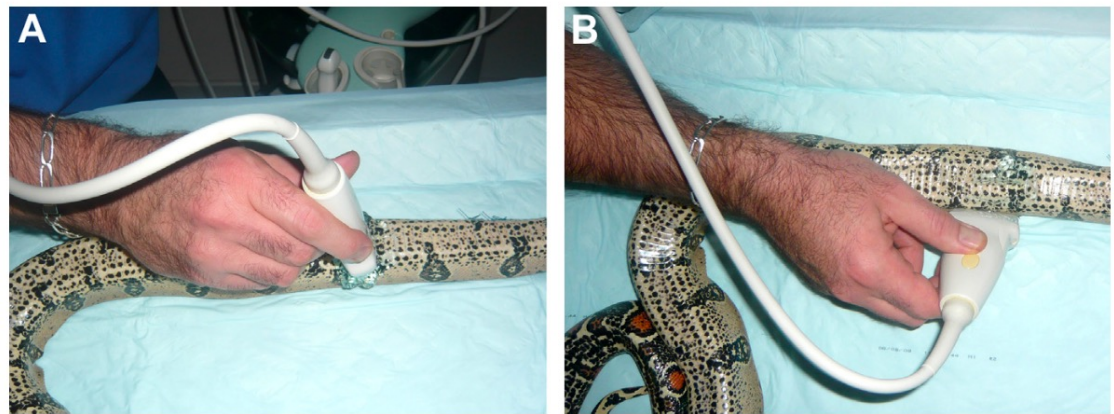
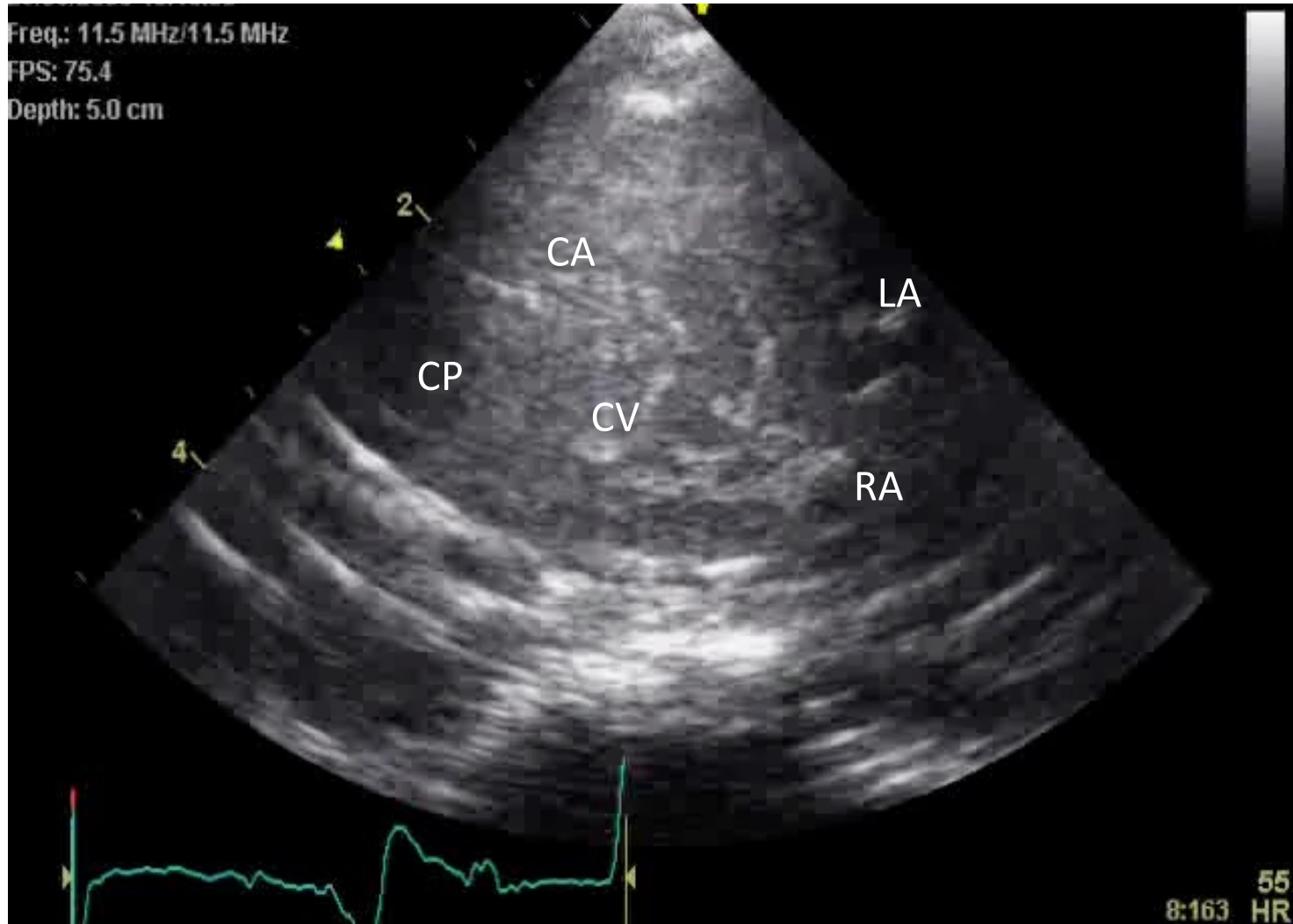


Fig. 15. Echocardiographic examination in a true red-tailed boa (*Boa constrictor constrictor*). (A) Ventral approach (short-axis section). (B) Left lateral approach (long-axis section).

Echocardiography



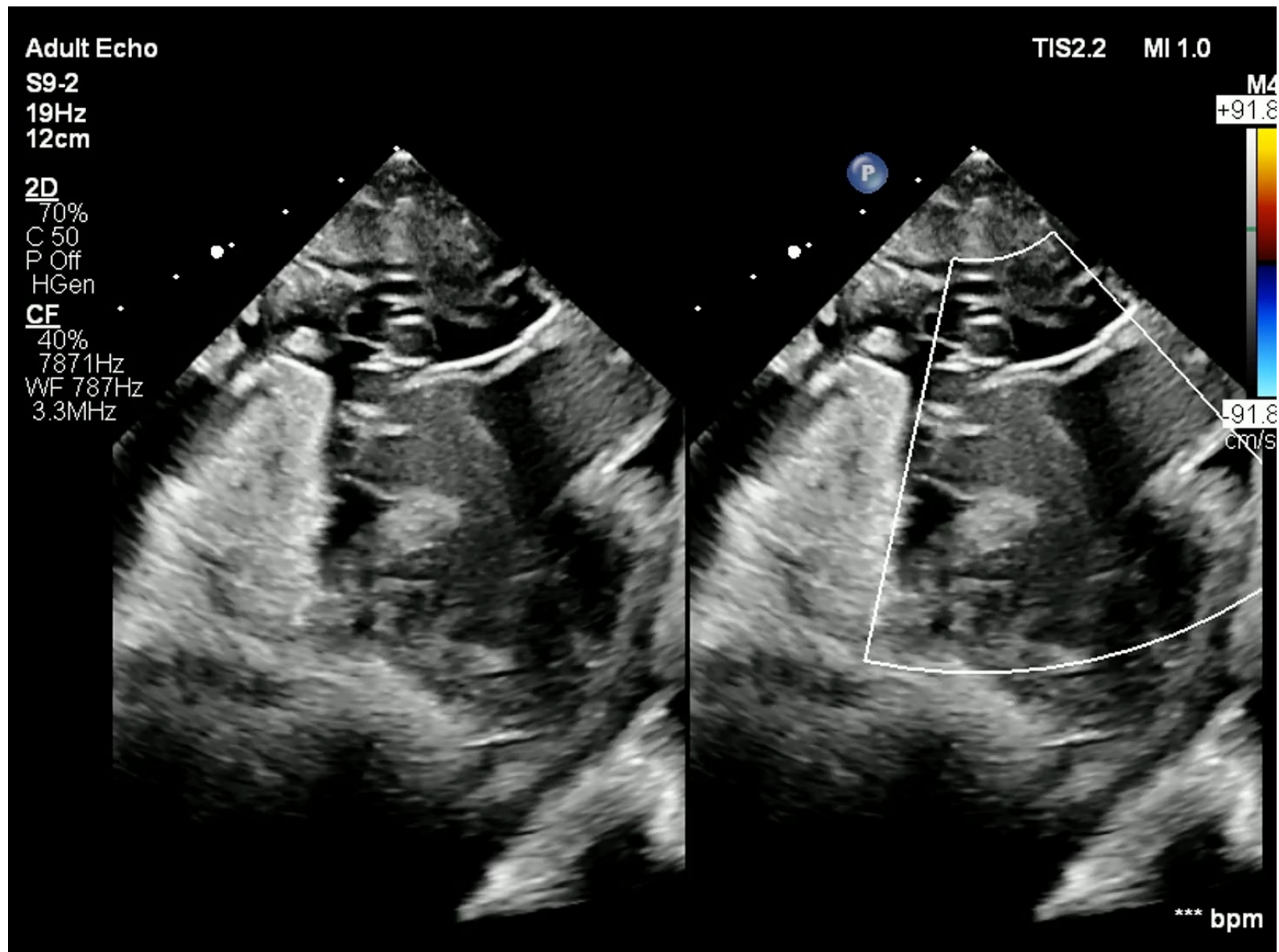
python

Echocardiography

- Cardiac motion/function
- Cardiac size
- Valve function
- Masses?
- Pericardial effusion
 - Normal in bearded dragons, red-tailed boas, red-eared sliders
- Operator-dependent
- Some reference values → some lizards, chelonians, boid snakes

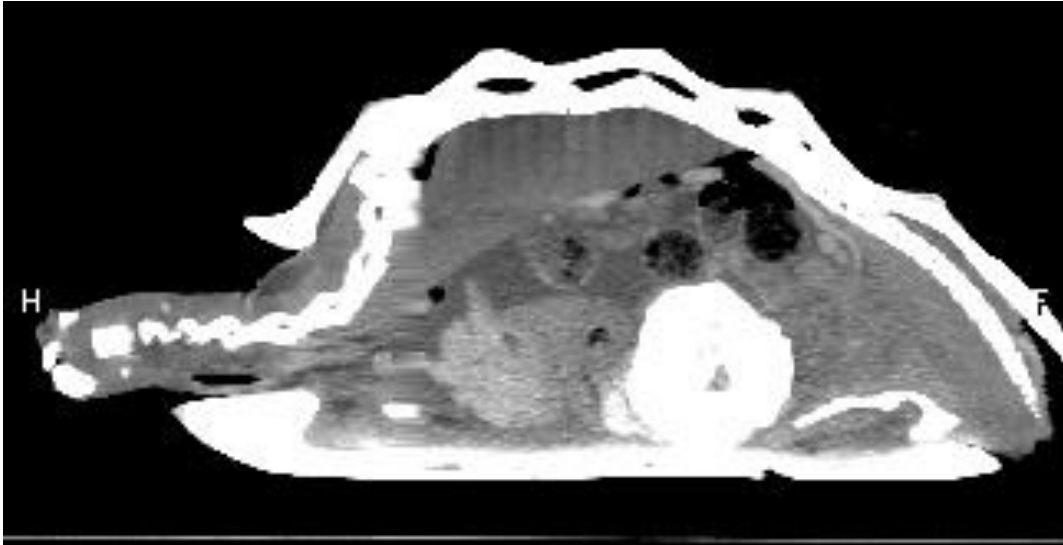


Chelonian



Advanced Imaging

- CT > MRI in chelonians
- \$\$\$\$, availability



Lecture Outline

- Anatomy and physiology
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- Diagnostics
- Etiologies of cardiovascular disease
- Treatment options

Congenital Heart Disease

- Rarely reported – might succumb to disease prior to hatching
- Consider in juveniles
- Aortic/subaortic stenosis in various species
- AV malformation reported (necropsy) in boa constrictor, python
- Atrial septal defect (Komodo dragon)



Nutrition-related CV disease

- Hypoalbuminemia
 - Malnutrition, liver disease
 - Edema, coelomic effusion, tachycardia
- Potassium derangements
 - Diet related or renal disease
 - Arrhythmogenic
- Gout (urate deposits in myocardium)
 - Excessive protein +/- dehydration



Nutrition-related CV disease

- Nutritional hyperparathyroidism
 - Common disease presentation in reptiles
 - ↓ Ca / ↑ P, Vit D3 deficiency
 - Lack of UVB light
- Hypocalcemia
 - Delayed myocardial depolarization, arrhythmias
 - ↓ pump function
 - Weakness, lethargy, tremors, seizures
- Vitamin E deficiency (white muscle disease)
 - Diet of obese rodents or oily fish
 - Tx: vitamin E and selenium

Nutrition-related CV disease

- Dystrophic mineralization of great vessels (Ao), myocardium
 - Tunica media
 - Increases afterload → heart failure
 - Ruptured Ao → Sudden death
 - Arrhythmias
- Excessive Vit D3 & Ca
 - Over supplemented, primate chows

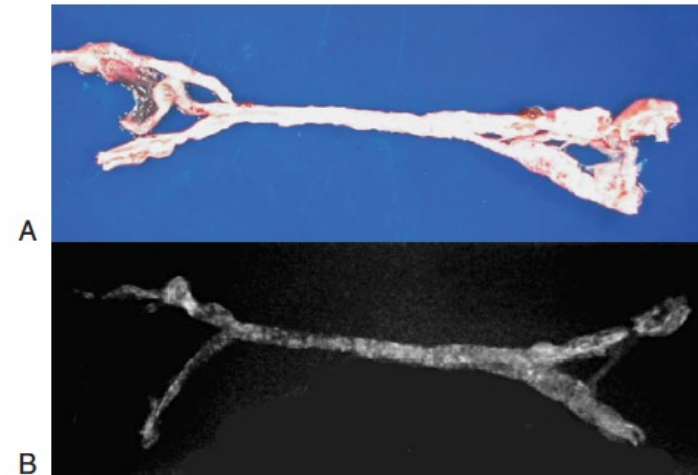
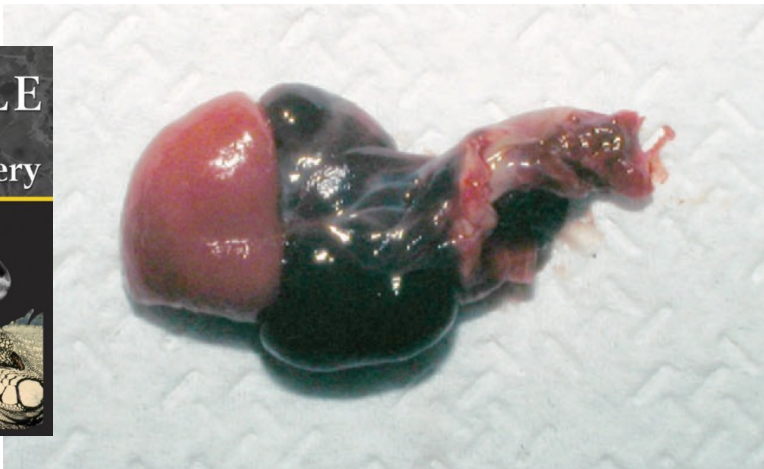
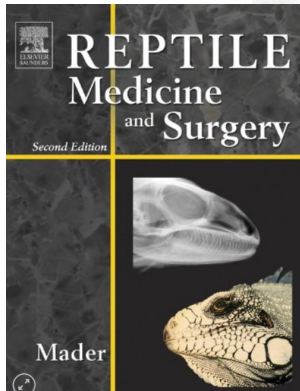


FIGURE 14-5 A, Resected aorta and large vessels from a Green Iguana (*Iguana iguana*). B, Radiograph of the same vessel in A. Note the calcified wall structure. (Photographs courtesy K. Rosenthal.)



Infectious CV disease

- CV system not usually primary target
- Bacterial infections
 - Gram neg bacteria → septicemia → endocarditis +/- thromboemboli
 - Myocarditis → myocardial dysfunction, arrhythmias
- Viral – rare case reports
- Parasites (can invade heart, great vessels)
 - Nematodes common
 - Dx: blood smear
 - Treatment: ivermectin (not chelonians)
 - Trematodes (Spirochid flukes) in chelonians
 - Dx: ELISA, blood smear
 - Treatment: praziquantel (efficacy unknown)

Acquired/primary CV disease

- Cardiomyopathies
 - Nutritional, hereditary (primary vs secondary)
 - Snakes > other reptiles
 - HCM, DCM, RCM
- Valvular disease
 - Typically endocarditic, degenerative possible
- Vascular disease
 - Vascular aneurysms, dissections → separation of muscular and intimal layers
- Pericardial disease
 - Infectious, visceral gout, hemopericardium (trauma, coagulopathy), neoplasia, manifestation of congestive heart failure
- Cardiac neoplasia
 - Various primary and metastatic cardiac neoplasias reported but rare

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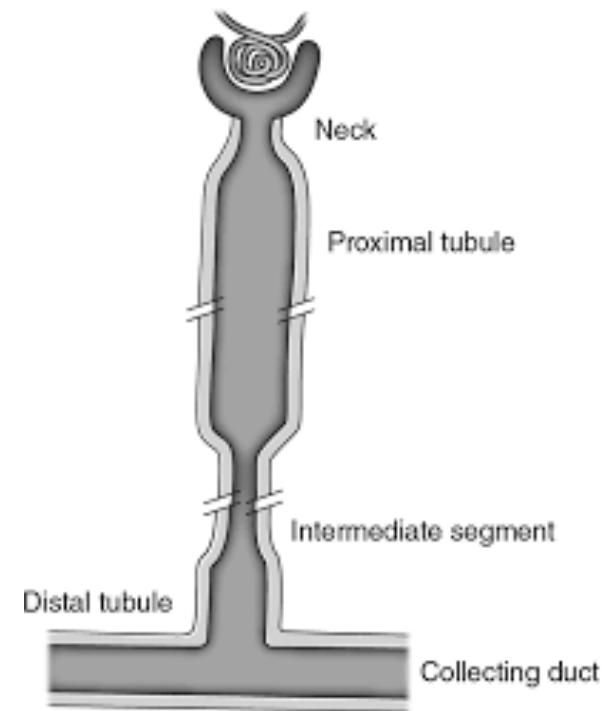
Clinical management

- ID and treat underlying cause
 - Nutrition, husbandry, etc.
 - Antibiotics
- Management:
 - Lower end of preferred optimal temperature zone
 - Reduce stress, separate from cage mates
 - Do not overfeed
 - Oxygen for dyspnea – may affect blood shunting

Medications

- Limited medication options
 - Diuretics
 - Furosemide – may be effective in some species
 - Diuretic effect shown in chelonians and water snakes
 - Mechanism: No loop of Henle, may work on Na/K pumps in bladder or terminal rectum/cloaca
 - Hydrochlorothiazide?
 - Pimobendan? (oral only) - has been used in lizards
 - ACE inhibitors? Spironolactone?
 - Anticholinergics, catecholamines
 - -centesis
- Often unsuccessful with long term prognosis being guarded to poor

Reptilian nephron



Questions?

